CSIT226 Human Computer Interaction Group Project – Assessment Guide

OVERVIEW OF THE PROJECT AND INSTRUCTIONS

Each lab class will be divided into workgroups of five (5) to six (6) students, by your tutor. Each group is to become a team to design the front-end (interfaces and interactions) of a system to solve an open complex information systems problem. In some weeks there will be time allocated in the labs for students to meet with their group members to work on the project.

Assessment for the project is based on the argument(s) developed for your system interfaces, the quality of the research used to support the interfaces developed, the ability to interpret what you have researched and the fluency of your written report.

You will be required to review each group members' performance through both self- and peer- assessment. The subject coordinator[s] will use this information to gain an understanding of individual participation and group contribution towards the project. This will also be used in determining individual marks for the group project (members will not necessarily be given the same marks).

COMPLEX COMPUTING PROBLEMS

This assessment has been created for groups to identify and design the user interfaces and interactions to solve a 'Complex Computing Problem' (based around the Seoul Accord[1]). Section D – Graduate Attributes of the Seoul Accord identify the role of a computing professional and D.4.1 discusses the concept of a 'Complex Computing Problem'.

POSSIBLE PROJECT DOMAINS

Initially, groups must choose a domain for their group project (from the list provided below). Groups then need to research and identify a specific issue in the domain that they want to develop a new system for (remember that you are only developing the user interfaces and interactions). Groups then need to research the kinds of systems that currently exist in their chosen domain and conduct an analysis of the issues with the systems and identify areas how they can be improved. This year all the domains come from the UN Sustainable Development Goals: https://sdgs.un.org/goals https://www.un.org/sustainabledevelopment/.

To start the project read the website and consider the 17 Goals[2]:

GOAL 1: No Poverty

GOAL 2: Zero Hunger

GOAL 3: Good Health and Well-being

GOAL 4: Quality Education

GOAL 5: Gender Equality

[1] See: https://www.seoulaccord.org/index.php
[2] See: https://www.undp.org/content/undp/en/home/sustainable-development-goals.html

GOAL 6: Clean Water and Sanitation

GOAL 7: Affordable and Clean Energy

GOAL 8: Decent Work and Economic Growth

GOAL 9: Industry, Innovation and Infrastructure

GOAL 10: Reduced Inequality

GOAL 11: Sustainable Cities and Communities

GOAL 12: Responsible Consumption and Production

GOAL 13: Climate Action

GOAL 14: Life Below Water

GOAL 15: Life on Land

GOAL 16: Peace and Justice Strong Institutions

GOAL 17: Partnerships to achieve the Goal

Review the SDGs in Action App (https://sdgsinaction.com/) and the information on the ActNow App (coming soon https://www.un.org/en/actnow/) "a mobile app in support of ActNow, offering a gamified experience for individuals to embark on a sustainable journey".

Groups are to choose a single UNSDG and build an application (interfaces and interactions) around making changes to support your chosen goal. Groups need to consider the target audience of the application and whether it would be for individuals or used within a particular industry. Consideration should be given if you are going to develop an e-commerce tool, educational tool (could be gamified), tracking tool etc. Consideration for some systems around having an administrative interface for updating content presented to users would also need to occur.

Remembering that there are marks for the overall creativity and innovation of your designs and project idea.

POSSIBLE TECHNOLOGIES AND APPLICATION ENVIRONMENTS

Groups must identify the technologies that will be used for the system being developed. This should only occur after a clear understanding of the problem domain and potential users/stakeholders. These could include (but are not limited to):

- Traditional computers (desktops and notebooks)
- Tablets
- Smartphones
- Smartwatches
- VR/AR devices
- IoT devices
- Artificial Intelligent systems

For the different technologies, groups would also need to consider the application environment that will be used in the design of the interfaces. For example, for traditional computers will a desktop application or a web-based application be designed?

Groups will need to consider design patterns and if any HIGs need to be followed.

Groups will also need to consider how the end-users would interact with the devices.

All design decisions must be grounded in the literature and prior system design research.

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TYPICAL REPORT STRUCTURE

A professional report *typically* has the following structure:

- Letter of transmittal – addressed to the recipient of the report.
- Title page
- Glossary / List of abbreviations
- Executive Summary
- Table of Contents (possible List of Figures / List of Tables)
- Introduction
- *{Body Text} this is split up under different headings depending on the nature and content of the report*
- Conclusion
- References
- Appendices

Note: It is expected that the report is created in a professional manner using the inbuilt features in word processor software (e.g. MS Word, Google Docs) for creating headings, figures and captions, sections etc.

PART A - OVERVIEW

Part A is designed for groups to show that they understand the domain that they are working with, potential users/stakeholders, develop personas and scenarios and initial system requirements.

Groups will initially have to conduct research about the domain that they will be working with and identify current systems or applications that are used. Groups will then need to identify the potential users (also stakeholders) and start to think about the problems that they could be facing. This would form the basis of the methods chosen for data gathering (e.g. interviews, focus groups, questionnaires).

POTENTIAL {BODY TEXT} FOR PART A

The following is one method to present the body of the report:

- Problem Domain Analysis
- Problem description / Solution Ideas
- Linked to your chosen UNSDG
- Type of overall application to be developed
- Current Systems Evaluation
  An evaluation of current systems that have similar features (these may not directly relate to the UNSDGs Could also use an analogous system as part of the evaluation focusing on features
- Stakeholder Analysis
- Users
- Primary
- Secondary
- Tertiary
- Non-users impacted by the system
- Persona[s]
- Scenario[s]
- Storyboard[s] presented along with detailed descriptions
- Initial Requirements (functional and non-functional)

PART B - OVERVIEW

Part B of the assessment is for you to demonstrate your creativity and innovation in coming up with your UI and UX for your chosen domain. There are numerous methods to demonstrate an iterative design approach and how to document this.

Groups should focus in detail on one or two scenarios of how the system would work allowing for greater depth in the details of a smaller aspect of the system rather than presenting hundreds of diagrams with limited explanation.

The focus should be about your group being able to justify your selection of design patterns and interaction sequences.

TYPES OF DIAGRAMS

The following are some of the types of diagrams that are expected in Part B:

- Storyboards
- Wireframes and 'wireflows'
- Lo-fi prototypes
- Hi-fi prototypes

When creating diagrams to present to a client on the user interface of the system visual representations are of high importance, UML diagrams are typically used within the project team and with some external clients that have a high level of technical knowledge are not required in CSIT226.

POTENTIAL {BODY TEXT} FOR PART B

The following is one method to present the body of the report (I don't suggest that you use these as exact headings):

- Content from Part A (in consideration of *feedback)*
- Interface development iterations
- Different designs presented – wireframe/workflow level
- Designs at low fidelity prototyping stage
- Two set of designs per group
- Discussion on the evaluation and acceptance of the design solutions
- Design iterations and selection of final design solution
- Final design solution modified to high fidelity stage ('near photo quality') • Overall design consideration and how the system will meet the requirements.
- How the system would be tested?

PART C - OVERVIEW

A Group video presentation which highlight your Problem Analysis, Solution Ideas, Design Process, UI/UX Interfaces that you have designed in Part B.

Discussing the key features, design patterns and consideration of any standards.

Assessment Description and Criteria

Length / Duration	2000 - 3000 words for Project Summary 6 - 15 minutes for Video